PassFrame tuition

Ratio & Proportion

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Write the ratio of 4 red beads to 6 blue beads in simplest form.
Method & answer: 4:6 = 2:3.

Standard: Share £60 in the ratio 2:3.
Method & answer: Total parts = 5. £60 ÷ 5 = £12 per part. 2 parts = £24, 3 parts = £36.

Advanced: A recipe for 4 people needs 200 g of rice. How much rice is needed for 10 people?
Method & answer: 200 g ÷ 4 = 50 g per person. 50 g × 10 = 500 g.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Write the ratio 6:8 in simplest form.

2. Write the ratio 10:15 in its simplest form. A) 1:2 B) 2:3 C) 5:10 D) 10:15

3. There are 5 cats and 10 dogs. Write the ratio of cats to dogs in simplest form.

4. Share £20 in the ratio 1:4. Select ALL statements that are TRUE: A) The shares are £4 and £16 B) The shares are £10 and £10 C) There are 5 parts in total D) The larger share is £16 E) There are 4 parts in total

5. Share 30 sweets in the ratio 2:3.

6. Write the ratio 12:18 in simplest form.

7. For every 2 red counters there are 3 blue counters. If there are 6 red counters, how many blue counters are there? A) 6 B) 8 C) 9 D) 12

8. A recipe for 2 people needs 100g of pasta. How much is needed for 4 people?

9. Write the ratio 9:6 in simplest form.

10. Share £40 in the ratio 3:5.

11. A map scale is 1 cm : 5 km. How many km does 3 cm represent?

12. Write the ratio 15:20 in its simplest form. A) 3:4 B) 3:5 C) 5:4 D) 1:5

13. There are 8 boys and 12 girls. Write the ratio of boys to girls in simplest form.

14. Share 24 marbles in the ratio 1:2. Select ALL statements that are TRUE: A) The shares are 8 and 16 B) The shares are 12 and 12 C) There are 3 parts in total D) The larger share is 16 E) There are 2 parts in total

15. A recipe for 3 people needs 150g of flour. How much is needed for 6 people?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Share £72 in the ratio 2:3:4.

6. Share 56 sweets in the ratio 3:4.

11. The ratio of cats to dogs at a shelter is 5:3. There are 40 animals in total. Select ALL statements that are TRUE: A) There are 25 cats B) There are 8 parts in total C) There are 15 dogs D) There are 20 cats E) 1 part is 5 animals

2. A recipe for 6 people needs 300g of rice. How much is needed for 8 people? A) 350g B) 375g C) 400g D) 450g

3. The ratio of red to blue paint is 3:5. If there are 15 litres of red, how much blue is there?

4. Simplify the ratio 24:36.

5. A class has a ratio of boys to girls of 3:4. There are 21 boys. Select ALL statements that are TRUE: A) There are 28 girls B) There are 7 parts in total C) 1 part is 7 pupils D) There are 24 girls E) 1 part is 3 pupils

7. A model car is built to a scale of 1:20. If the model is 8 cm long, how long is the real car? A) 28cm B) 100cm C) 160cm D) 180cm

8. For a fruit punch, the ratio of juice to lemonade is 2:5. If 10 litres of juice are used, how much lemonade is needed?

9. Share £90 in the ratio 4:5.

10. A recipe for 5 people needs 250 ml of milk. How much is needed for 12 people?

12. Simplify the ratio 45:30. A) 3:2 B) 2:3 C) 9:6 D) 15:10

13. A paint mix uses blue and white in the ratio 1:4. How much white paint is needed for 3 litres of blue?

14. Share 100 counters in the ratio 2:3.

15. A scale drawing uses 1 cm to represent 4 m. How many cm represent 24 m?

Standard / Hard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A fruit drink is made from orange and mango juice in the ratio 3:2. If a jug contains 800ml of drink in total, select ALL statements that are TRUE: A) There are 5 parts in total B) There is 480ml of orange juice C) There is 320ml of mango juice D) There is 400ml of orange juice E) Each part is 160ml

2. A recipe for 4 people needs 240g of pasta and 2 jars of sauce. If you make the recipe for 10 people, how much pasta and how many jars of sauce are needed?

3. The ratio of teachers to pupils on a school trip is 1:8. If there are 96 pupils, how many teachers are needed? A) 8 B) 12 C) 10 D) 14

4. A sum of money is shared between Amy and Ben in the ratio 3:5. Ben receives £45 more than Amy. How much does Amy receive?

5. A scale model of a building is built at 1:50. The real building is 25 m tall. How tall is the model, in cm?

6. In a bag of counters, the ratio of red to green to yellow is 2:3:5. If there are 20 red counters, how many counters are there in total?

7. A recipe uses flour and sugar in the ratio 5:2. If 350g of flour is used, how much sugar is needed, and what is the total weight of flour and sugar?

8. Two numbers are in the ratio 4:7. Their sum is 99. What is the larger number? A) 36 B) 44 C) 55 D) 63

9. The ratio of adults to children on a coach is 2:5. There are 14 adults. How many people are on the coach in total? A) 35 B) 42 C) 49 D) 56

10. A car travels 180 miles using 20 litres of fuel. Select ALL statements that are TRUE at the same rate: A) It travels 9 miles per litre B) On 35 litres it can travel 315 miles C) On

11. The ratio of a rectangle's length to width is 5:2. If the width is 8 cm, what is the perimeter of the rectangle?

12. A photo is enlarged in the ratio 3:8. If the original width is 9 cm and height is 6 cm, what are the new dimensions?

13. A recipe for 4 people uses 3 eggs. How many eggs are needed for 10 people, rounding up to the nearest whole egg?

14. A sum of £160 is shared in the ratio 1:3:4. What is the difference between the largest and smallest shares?

15. A map has a scale of 1:25,000. Two towns are 8 cm apart on the map. How far apart are they in real life, in km?

35 litres it can travel 300 miles D) It travels 10 miles per litre E) On 10 litres it can travel 90 miles